

AIR POLLUTION AND CPCB STANDARDS

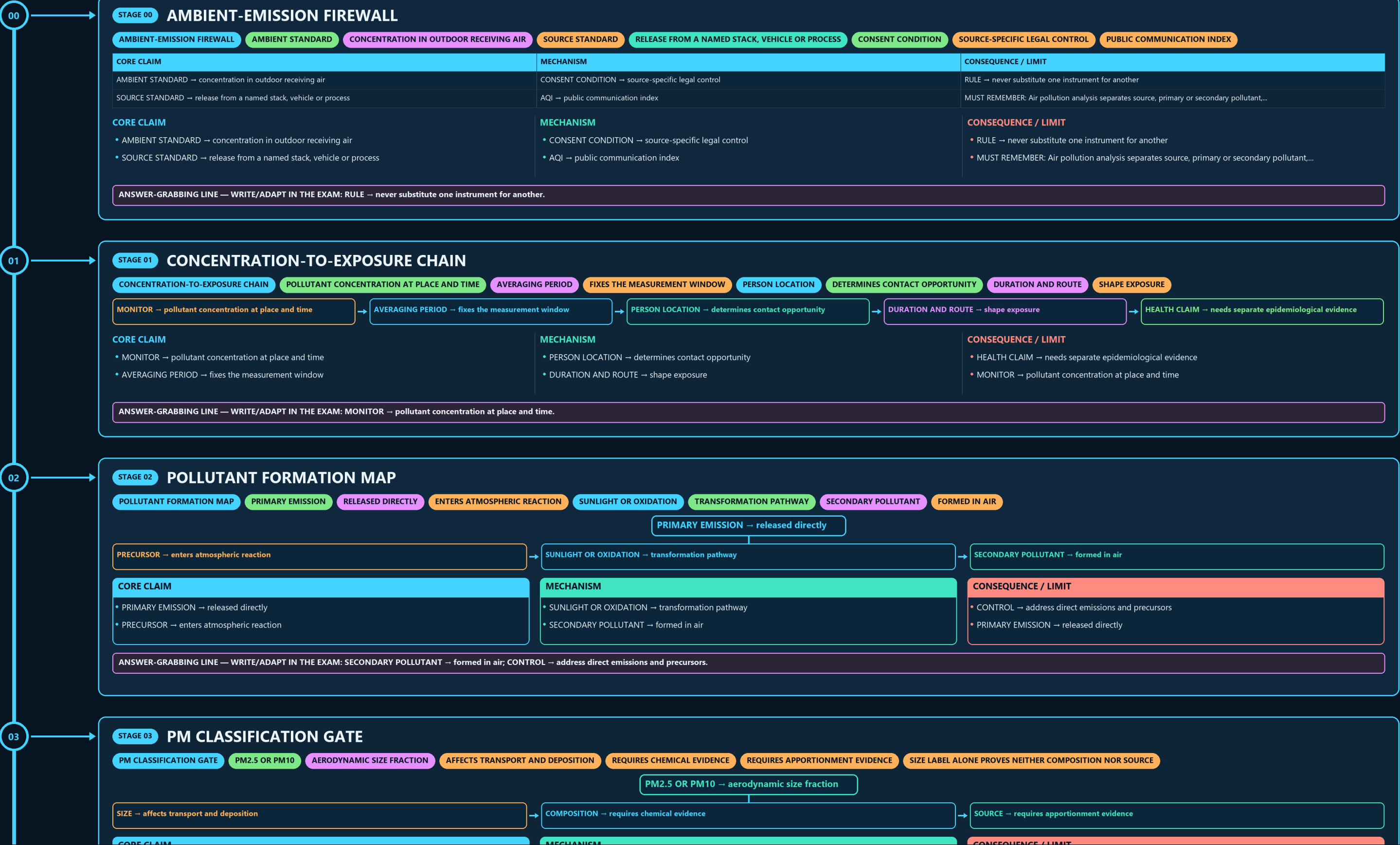
AMBIENT-EMISSION FIREWALL → CONCENTRATION-TO-EXPOSURE CHAIN → POLLUTANT FORMATION MAP → PM CLASSIFICATION GATE → SMOG AND ACID-DEPOSITION MATRIX → NAAQS DATA GATE

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SECONDARY POLLUTANT → formed in air; CONTROL → address direct emissions and precursors.

03

STAGE 03 PM CLASSIFICATION GATE

PM CLASSIFICATION GATE PM2.5 OR PM10 AERODYNAMIC SIZE FRACTION AFFECTS TRANSPORT AND DEPOSITION REQUIRES CHEMICAL EVIDENCE REQUIRES APPORTIONMENT EVIDENCE SIZE LABEL ALONE PROVES NEITHER COMPOSITION NOR SOURCE

PM2.5 OR PM10 → aerodynamic size fraction

SIZE → affects transport and deposition

COMPOSITION → requires chemical evidence

SOURCE → requires apportionment evidence

CORE CLAIM

- PM2.5 OR PM10 → aerodynamic size fraction
- SIZE → affects transport and deposition

MECHANISM

- COMPOSITION → requires chemical evidence
- SOURCE → requires apportionment evidence

CONSEQUENCE / LIMIT

- TRAP → size label alone proves neither composition nor source
- PM2.5 OR PM10 → aerodynamic size fraction

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → size label alone proves neither composition nor source.

04

STAGE 04 SMOG AND ACID-DEPOSITION MATRIX

SMOG AND ACID-DEPOSITION MATRIX PRECURSOR EMISSIONS PLUS SUNLIGHT-DRIVEN CHEMISTRY SURFACE OZONE SECONDARY OXIDANT ACID DEPOSITION SULPHUR AND NITROGEN PRECURSOR PATHWAY WET OR DRY DEPOSITION ROUTES

CORE CLAIM

SMOG → precursor emissions plus sunlight-driven chemistry
SURFACE OZONE → secondary oxidant

MECHANISM

ACID DEPOSITION → sulphur and nitrogen precursor pathway
WET OR DRY → deposition routes

CONSEQUENCE / LIMIT

RULE → precursor is not the final atmospheric product
SMOG → precursor emissions plus sunlight-driven chemistry

CORE CLAIM

- SMOG → precursor emissions plus sunlight-driven chemistry
- SURFACE OZONE → secondary oxidant

MECHANISM

- ACID DEPOSITION → sulphur and nitrogen precursor pathway
- WET OR DRY → deposition routes

CONSEQUENCE / LIMIT

- RULE → precursor is not the final atmospheric product
- SMOG → precursor emissions plus sunlight-driven chemistry

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → precursor is not the final atmospheric product.

05

STAGE 05 NAAQS DATA GATE

NAAQS DATA GATE IDENTIFY EXACT SUBSTANCE OR FRACTION PRESERVE THE NOTIFIED UNIT AVERAGING PERIOD ANNUAL, DAILY OR SHORTER ONLY IF SOURCED AREA AND COMPLIANCE RULE READ NOTIFICATION NO TABLE

POLLUTANT → identify exact substance or fraction

UNIT → preserve the notified unit

AVERAGING PERIOD → annual, daily or shorter only if sourced

AREA AND COMPLIANCE RULE → read notification

CORE CLAIM

- POLLUTANT → identify exact substance or fraction
- UNIT → preserve the notified unit

MECHANISM

- AVERAGING PERIOD → annual, daily or shorter only if sourced
- AREA AND COMPLIANCE RULE → read notification

CONSEQUENCE / LIMIT

- NO TABLE → no invented limit
- POLLUTANT → identify exact substance or fraction

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AVERAGING PERIOD → annual, daily or shorter only if sourced.

06

STAGE 06 AQI STATUTORY BOUNDARY

AQI STATUTORY BOUNDARY MONITORED POLLUTANT CONCENTRATIONS PUBLIC-FACING CATEGORY NOTIFIED AMBIENT STANDARD SOURCE LIMIT SEPARATE EMISSION CONTROL AQI CATEGORY IS NOT EITHER STANDARD TABLE

CORE CLAIM

INPUT → monitored pollutant concentrations
OUTPUT → public-facing category

MECHANISM

NAAQS → notified ambient standard
SOURCE LIMIT → separate emission control

CONSEQUENCE / LIMIT

VERDICT → AQI category is not either standard table
INPUT → monitored pollutant concentrations

CORE CLAIM

- INPUT → monitored pollutant concentrations
- OUTPUT → public-facing category

MECHANISM

- NAAQS → notified ambient standard
- SOURCE LIMIT → separate emission control

CONSEQUENCE / LIMIT

- VERDICT → AQI category is not either standard table
- INPUT → monitored pollutant concentrations

06

STAGE 06 AQI STATUTORY BOUNDARY

AQI STATUTORY BOUNDARY MONITORED POLLUTANT CONCENTRATIONS PUBLIC-FACING CATEGORY NOTIFIED AMBIENT STANDARD SOURCE LIMIT SEPARATE EMISSION CONTROL AQI CATEGORY IS NOT EITHER STANDARD TABLE

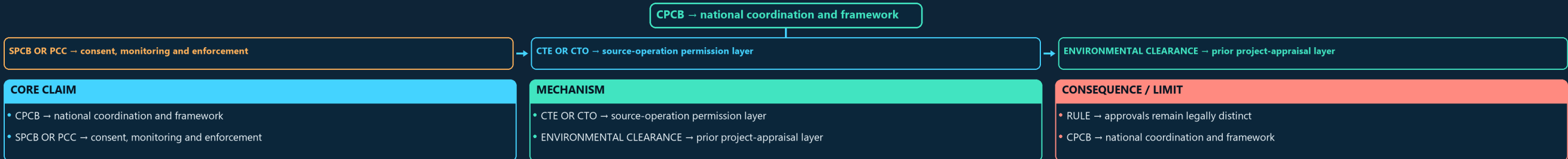
CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
INPUT → monitored pollutant concentrations	NAAQS → notified ambient standard	VERDICT → AQI category is not either standard table
OUTPUT → public-facing category	SOURCE LIMIT → separate emission control	INPUT → monitored pollutant concentrations
CORE CLAIM • INPUT → monitored pollutant concentrations • OUTPUT → public-facing category	MECHANISM • NAAQS → notified ambient standard • SOURCE LIMIT → separate emission control	CONSEQUENCE / LIMIT • VERDICT → AQI category is not either standard table • INPUT → monitored pollutant concentrations

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → AQI category is not either standard table.

07

STAGE 07 INSTITUTION AND CLEARANCE LADDER

INSTITUTION AND CLEARANCE LADDER NATIONAL COORDINATION AND FRAMEWORK SPCB OR PCC CONSENT, MONITORING AND ENFORCEMENT CTE OR CTO SOURCE-OPERATION PERMISSION LAYER ENVIRONMENTAL CLEARANCE PRIOR PROJECT-APPRAISAL LAYER

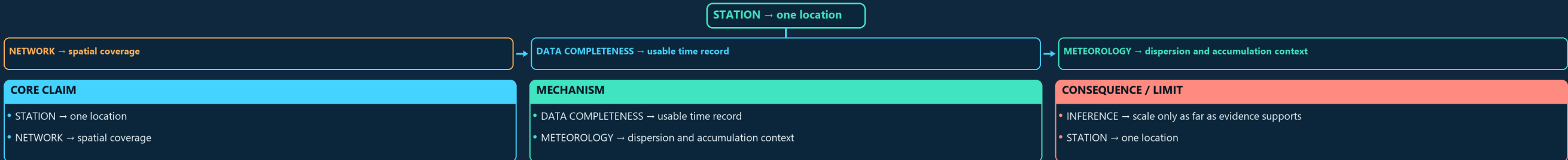


ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SPCB OR PCC → consent, monitoring and enforcement.

08

STAGE 08 MONITORING EVIDENCE LADDER

MONITORING EVIDENCE LADDER ONE LOCATION SPATIAL COVERAGE DATA COMPLETENESS USABLE TIME RECORD DISPERSION AND ACCUMULATION CONTEXT SCALE ONLY AS FAR AS EVIDENCE SUPPORTS



ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INFERENCE → scale only as far as evidence supports.

09

STAGE 09 SOURCE-APPORTIONMENT CUBE

SOURCE-APPORTIONMENT CUBE SOURCE MIX CHANGES BY LOCATION ACTIVITY AND METEOROLOGY CHANGE PM, OZONE AND GASES HAVE DIFFERENT PATHWAYS MODEL AND SAMPLING ASSUMPTIONS MATTER NO UNIVERSAL SHARE DATE AND SCOPE EVERY ESTIMATE CLOSE DISTINCTION

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
CITY → source mix changes by location	POLLUTANT → PM, ozone and gases have different pathways	NO UNIVERSAL SHARE → date and scope every estimate
SEASON → activity and meteorology change	METHOD → model and sampling assumptions matter	CLOSE DISTINCTION: Emission is not ambient concentration or exposure, AQI is not an...
CORE CLAIM • CITY → source mix changes by location • SEASON → activity and meteorology change	MECHANISM • POLLUTANT → PM, ozone and gases have different pathways • METHOD → model and sampling assumptions matter	CONSEQUENCE / LIMIT • NO UNIVERSAL SHARE → date and scope every estimate • CLOSE DISTINCTION: Emission is not ambient concentration or exposure, AQI is not an...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Emission is not ambient concentration or exposure, AQI is not an.

09

STAGE 09 SOURCE-APPORTIONMENT CUBE

SOURCE-APPORTIONMENT CUBE SOURCE MIX CHANGES BY LOCATION ACTIVITY AND METEOROLOGY CHANGE PM, OZONE AND GASES HAVE DIFFERENT PATHWAYS MODEL AND SAMPLING ASSUMPTIONS MATTER NO UNIVERSAL SHARE DATE AND SCOPE EVERY ESTIMATE CLOSE DISTINCTION

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CORE CLAIM - CITY → source mix changes by location - SEASON → activity and meteorology change	MECHANISM - POLLUTANT → PM, ozone and gases have different pathways - METHOD → model and sampling assumptions matter	CONSEQUENCE / LIMIT - NO UNIVERSAL SHARE → date and scope every estimate - CLOSE DISTINCTION: Emission is not ambient concentration or exposure, AQI is not an...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Emission is not ambient concentration or exposure, AQI is not an.

10

STAGE 10 PROGRAMME-RESPONSE ARCHITECTURE

PROGRAMME-RESPONSE ARCHITECTURE LONG-RUN STRUCTURAL PLANNING STANDARDS-BASED CLASSIFICATION OVER A PERIOD EPISODIC GRADED RESPONSE STATUTORY DELHI-NCR REGIONAL COORDINATION NO SUBSTITUTION EACH SERVES A DIFFERENT FUNCTION EVIDENCE LIMIT

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
NCAP → long-run structural planning	GRAP → episodic graded response	NO SUBSTITUTION → each serves a different function
NON-ATTAINMENT → standards-based classification over a period	CAQM → statutory Delhi-NCR regional coordination	EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State pollutant, averaging time, unit,...
CORE CLAIM - NCAP → long-run structural planning - NON-ATTAINMENT → standards-based classification over a period	MECHANISM - GRAP → episodic graded response - CAQM → statutory Delhi-NCR regional coordination	CONSEQUENCE / LIMIT - NO SUBSTITUTION → each serves a different function - EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State pollutant, averaging time, unit,...

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State pollutant, averaging time, unit,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State pollutant, averaging time, unit,.

11

STAGE 11 EVIDENCE AND ANSWER SPINE

EVIDENCE AND ANSWER SPINE AMBIENT, EMISSION, CONCENTRATION AND EXPOSURE SOURCE, PRECURSOR, ATMOSPHERIC FORMATION AND RECEPTOR NAAQS, AQI, CONSENT, NCAP, GRAP AND CAQM STATION, AVERAGING PERIOD AND APPORTIONMENT BOUNDARY

TARGET, ATTAINMENT, CURRENT DATA AND PYQ KEY STAY DISTINCT

DEFINE → ambient, emission, concentration and exposure			→	TRACE → source, precursor, atmospheric formation and receptor			→	GOVERN → NAAQS, AQI, consent, NCAP, GRAP and CAQM			→	MEASURE → station, averaging period and apportionment boundary			→	AUDIT → target, attainment, current data and PYQ key stay distinct		
CORE CLAIM						MECHANISM						CONSEQUENCE / LIMIT						
• DEFINE → ambient, emission, concentration and exposure • TRACE → source, precursor, atmospheric formation and receptor						• GOVERN → NAAQS, AQI, consent, NCAP, GRAP and CAQM • MEASURE → station, averaging period and apportionment boundary						• AUDIT → target, attainment, current data and PYQ key stay distinct • DEFINE → ambient, emission, concentration and exposure						

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AUDIT → target, attainment, current data and PYQ key stay distinct.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT EFFECTIVE AIR-POLLUTION POLICY DEPENDS ON ACCURATE SOURCE APPORTIONMENT (SCIENTIFICALLY) COMMON ANALYTICAL ERROR TO AVOID TREATING CROP-RESIDUE BURNING AS THE DOMINANT

NCAP SET CITY-SPECIFIC PARTICULATE-MATTER (PM2.5/PM10) REDUCTION TARGETS FOR IDENTIFIED IMPLEMENTATION CRITIQUE PROGRESS TRACKING UNDER NCAP HAS FACED CHALLENGES BALANCED FRAMING

OPTIONAL SOURCE DEPTH - CAUTION: Effective air-pollution policy depends on accurate source apportionment (scientifically) - CAUTION: Common analytical error to avoid: treating crop-residue burning as the dominant, - NCAP set city-specific particulate-matter (PM2.5/PM10) reduction targets for identified - CAUTION: Implementation critique: progress tracking under NCAP has faced challenges	ADVANCED DEBATE - CAUTION: Balanced framing: NCAP represents a scientifically sound target-setting framework; - CAUTION: Real-time AQI readings depend on monitoring-station density and placement; sparse - CAUTION: NAAQS permissible limits are themselves periodically reviewed against evolving health - CAUTION: Attributing health outcomes (mortality, morbidity) precisely to a specific pollution	LEGACY / NUANCE - The Commission for Air Quality Management in NCR and Adjoining Areas Act, 2021 - GRAP is a reactive, severity-triggered emergency-response mechanism; NCAP is a - Source apportionment studies show the relative contribution of crop-residue burning to - CAQM was established partly because prior coordination on Delhi-NCR air pollution had
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OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.